

31. Stacked Bar Plot with Error Bars

Question

Write a Python program to create a stacked bar plot with error bars.

Aim

To create a stacked bar plot with error bars using Python and Matplotlib.

Code import matplotlib.pyplot as plt import numpy as np

```
# Data
```

```
groups = ['A', 'B', 'C', 'D']
```

```
part1 = [20, 30, 25, 35] part2
```

```
= [15, 20, 30, 25]
```

```
# Error values error1
```

```
= [2, 3, 2, 4] error2 =
```

```
[1, 2, 3, 2]
```

```
# Create stacked bar plot plt.bar(groups, part1, yerr=error1, capsize=5,
```

```
label='Part 1') plt.bar(groups, part2, bottom=part1, yerr=error2, capsize=5,
```

```
label='Part 2')
```

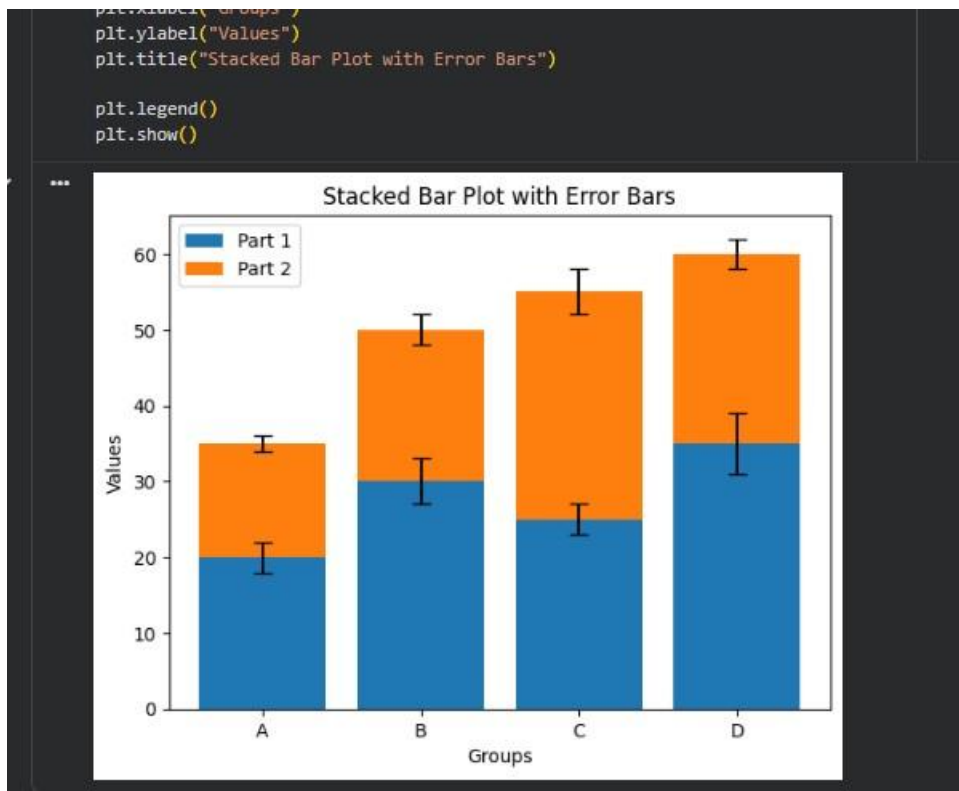
```
# Labels and title plt.xlabel("Groups")
```

```
plt.ylabel("Values") plt.title("Stacked Bar
```

```
Plot with Error Bars")
```

```
plt.legend()
```

```
plt.show() Output:
```



32. Scatter Graph Using Random Distribution

Question

Write a Python program to draw a scatter graph taking a random distribution in X and Y and plotted against each other.

Aim

To create a scatter graph using randomly generated X and Y values. **Code**

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
# Generate random X and Y values np.random.seed(10)
```

```
x = np.random.rand(50)
```

```
y = np.random.rand(50)
```

```
Create scatter plot
```

```
plt.scatter(x, y)
```

```
# Labels and title plt.xlabel("X Values")  
plt.ylabel("Y Values") plt.title("Scatter Plot of  
Random X and Y Values")
```

plt.show() **Output:**

